CO-2 - AT-1

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Course :- Computer Vision

Code : ITA0502

Given

$r = [25, 60, 95, 130, 165, 200, 235]$

Intensity band = (90 — 170)

a)

$$S = \begin{cases} 255, & r \geq 128 \\ 0, & r < 128 \end{cases}$$

I/P	25	60	95	130	165	200	235
O/P	0	0	0	255	255	255	255

Ans = (O/P) = [0, 0, 255, 255, 255, 255]

b)

$$S = \begin{cases} 255, & 90 \leq r \leq 170 \\ r, & \text{otherwise} \end{cases}$$

O/P = [25, 60, 255, 255, 255, 200, 235]

c)

No. of highlighte pixels = 3

Total = 7

$$\% = \frac{3}{7} \times 100$$

$$= 42.86\%$$

d)

Mean intensity.

$$\bar{r} = \frac{25+60+95+130+165+200+235}{7}$$

$$\boxed{\bar{r} = 130}$$

Intensity level slicing is more appropriate. isolating solder joints because of high gray pixels

3)

Given: $r_1 = 70$, $s_1 = 20$, $L = 256$

$r_2 = 140$, $s_2 = 230$

$r = [20, 70, 90, 110, 140, 200]$

a) $s = ar + b$

Segment 1: $0 \le r \le 70$

$$m_1 = \frac{20-0}{70-0} = \frac{2}{7} = 0.2857$$

$$\boxed{s = 0.2857r}$$

Segment 2: $70 \le r \le 140$

$$m_2 = \frac{230-20}{140-70} = 3 \Rightarrow \boxed{s = 3r - 190}$$

Segment 2: $140 \le r \le 255$

$$m_3 = \frac{255-230}{255-140} = \frac{25}{115} = 0.2174$$

$$\boxed{s = 0.2174r + 199.5652}$$

b) Calculated Pixel Values:

I/P	30	70	90	110	140	20
O/P	8.57	20	80	140	230	243

O/P = [8.57, 20, 80, 140, 230, 243.04]

c) Band Width:

$W_{original} = 140 - 70 = 70$

$W_{transformed} = 230 - 20 = 210$

$G = \frac{210}{70} = 3 \Rightarrow \boxed{G = 3}$

Compression	Slope	Effect
0-70	0.2857	Compression
70-140	3	Expansion
140-255	0.2174	Compression

c) Conclusion:

The transformation improves the visibility of diagnostic details in the 70-140 intensity range because contrast is expanded by factor of 3.

Given neighbourhood:

$$\begin{bmatrix} 52 & 55 & 58 \\ 54 & 255 & 57 \\ 53 & 56 & 59 \end{bmatrix}$$

255 → noise impulse

a) Mean = $\frac{\text{Total pixels}}{\text{no of pixels}}$

$$= \frac{52+55+58+54+255+57+53+56+59}{9}$$

mean = 77.67

b) Weighted - Avg Filter:

$$\frac{1}{16}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

Weighted Sum = 1688

$$S = \frac{1688}{16}$$

S = 105.5

c) Median Filter:

A·O = [52, 53, 54, 55, 56, 57, 58, 59, 255]

median = 56 (noise removed)

d) Absolute error (noise free referenced)

$$R = \frac{52+55+58+54+57+53+56+57}{8}$$

$$R = \frac{444}{8} = 55.5$$

Filter	O/P	Absolute error
Mean	77.67	22.17
wgt avg	105.5	50
Median	56	0.5

e) Conclusion:

The median filter is the most approroate for removing salt & pepper noise while preserving edges. It removes extreme values.

4)

Given Image region

$$\begin{bmatrix} 10 & 10 & 10 \\ 10 & 10 & 90 \\ 10 & 90 & 90 \end{bmatrix}$$

a) G_x $\begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$

By Multiplying $G_x = (-10+10)+(-20+180)+(-10+90)$

$G_x = 240$

Gy mask $\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$

$G_y = (-10 - 20 - 10) + (10 + 180 + 90) \rightarrow 240$

b) Gradient Magnitude $(M) = \sqrt{G_x^2 + G_y^2}$

$M = \sqrt{240^2 + 240^2}$

$\boxed{M = 339.41}$

$M_a = |G_x| + |G_y|$

$\boxed{M_a = 480}$

% error $(E) = \frac{M_a - M}{M} \times 100$

$= 41.42\%$

c) Gradient direction $(\alpha) = \tan^{-1}\left(\frac{G_y}{G_x}\right) = \tan^{-1}\left(\frac{240}{240}\right)$

$\boxed{\alpha = 45^\circ}$

d) Prewitt operation

G_x mask $= \begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix}$, $G_x = (-10 + 10) + (-10 + 90) + (-10 + 90)$

$G_x = 160$

G_y mask $= \begin{bmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix} = (-10 - 10 - 10) + (10 + 90 + 90)$

$G_y = 160$

$\boxed{M = \sqrt{G_x^2 + G_y^2} = 226.27}$

$M_a = \frac{320 - 226.27}{226.27} \times 100$

$\boxed{E = 41.42\%}$

e)

Operator	M_a	Threshold	Edge
Sobel	339.41	250	Yes
Prewitt	226.27	250	No.

Conclusion

Sobel detects the edge better than prewitt

5) Given

Intensity	0	1	2	3	4	5	6
Frequency	3	6	4	1	1	5	8

$N = 3+6+4+1+1+5+8+4 = 32$

a) Mean Intensity $(M_G) = \frac{\sum r_k n_k}{N}$

$= \frac{117}{32} \Rightarrow \boxed{m_G = 3.65625}$

b) Iterative global

$T_0 = m_G = 3.65625$

C_1 : pixels intensity $\leq T$

C_2 : " " $> T$

Iteration 1:

$m_1 = \frac{0(3)+1(6)+2(4)+3(1)}{3+6+4+1} = 1.2143$

$m_2 = \frac{4(1)+5(5)+6(8)+7(4)}{1+5+8+4} = \frac{105}{18}$

$= 5.8333$

$T_1 = \frac{m_1+m_2}{2}$

$T_1 = \frac{1.2143+5.833}{2}$

$\boxed{T_1 = 3.5238}$

$\Delta T = 0.13245 \quad (< 0.5)$

$\therefore \boxed{T = 3.5238}$

c) Otsu's b/w Class Variance

$\sigma_B^2(k) = P_1(k) P_2(k) [m_1(k) - m_2(k)]^2$

For $k = 2$

$P_1 = \frac{13}{32}, \quad m_1 = \frac{14}{13} = 1.0769$

For $k=2$

Class C_1: levels 0-2

$P_1 = \frac{13}{32}$, $m_1 = \frac{14}{13} = 1.0769$

C_2, levels 3-7

$P_2 = \frac{19}{32}$, $m_2 = \frac{103}{19} = 5.4211$

$\sigma_B^2(2) = \frac{13}{32} \cdot \frac{19}{32} (1.0769 - 5.4211)^2$

$\boxed{\sigma_B^2(2) \approx 4.49}$

At $K=3$,

$\sigma_B^2(3) = \frac{14}{32} \cdot \frac{18}{32} (1.2143 - 5.8333)^2$

$\boxed{\sigma_B^2(3) \approx 5.31}$

At $K=4$,

C_1 = $P_1 = \frac{15}{32}$, $m_1 = \frac{21}{15} = 1.4$

C_2 $P_2 = \frac{17}{32}$, $m_2 = \frac{103}{17} = 6.0588$

$\sigma_B^2(4) = \frac{15}{32} \cdot \frac{17}{32} (1.4 - 6.0588)^2$

$\boxed{\sigma_B^2(4) \approx 5.06}$

d) Optimum Otsu Threshold.

Threshold	class Variance
2	4.49
3	5.31
4	5.06

$\Rightarrow$ Max σ at $K=3$

$\therefore \boxed{K^* = 3}$

$S = 0, \quad r \le 3$

$\quad\quad 1, \quad r > 3$

I/P	0	1	2	3	4	5	6	7
Pixel	6	0	0	0	1	1	1	1

$N_{object} = 1 + 5 + 8 + 4 = 18$

% classified $= \frac{18}{32} \times 100 = 56.25\%$

c) Comparision & Conclusions

Method	Threshold
Iterative	3.5238
Otsu's	3

Both methods produce similar thresholds & divide the image into dark & bright. Otsu's method selects the threshold that maximizes btw class & variance